

An Integrated AI-Powered Interview Simulation and Proctoring Platform with Biometric Identity Verification, Multi-Modal Behavioural Analysis, and Adaptive Question Generation

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Abstract

Remote hiring processes present critical challenges related to candidate identity fraud, real-time coaching assistance, and the objective assessment of behavioural competencies. This paper presents an integrated AI-powered interview simulation and proctoring platform that addresses these challenges through a comprehensive multi-modal pipeline. A biometric pre-interview identity gate employs ArcFace deep face recognition combined with frame-variance and inter-frame mean squared error anti-spoofing heuristics to authenticate each candidate prior to session commencement. During the interview, a continuous anti-handoff monitoring mechanism verifies candidate identity on every incoming frame at a stricter cosine similarity threshold, flagging sessions after three consecutive mismatches. A real-time, asynchronous, multi-model behavioural proctoring pipeline runs in parallel, covering gaze direction via MediaPipe Face Mesh, emotional state via a VGG-based classifier, unauthorised face presence via Haar Cascade, and mobile phone detection via a custom YOLO model. Speech processing employs Wav2Vec 2.0 for tone analysis and OpenAI Whisper for speech-to-text transcription. Adaptive interview questions are generated dynamically by a DeepSeek large language model conditioned on candidate CVs and session context. The platform serves Candidate, HR Manager, and Administrator roles through a Flutter Web application and a dedicated Flutter Windows desktop client, communicating over a WebSocket bridge between a Node.js proxy and a Python FastAPI AI backend. Comparative analysis against five recent related systems demonstrates that the proposed platform is the only work to simultaneously address the full spectrum of remote interview integrity, analytics, and workflow requirements.

Keywords — interview simulation, biometric verification, ArcFace, anti-spoofing, gaze tracking, emotion recognition, large language model, real-time proctoring, WebSocket, Flutter, DeepSeek, Whisper, Wav2Vec.

I. INTRODUCTION

The global shift toward remote hiring has accelerated dramatically in recent years, with a significant proportion of recruitment processes now conducted entirely online. While remote interviews offer substantial logistical advantages, they introduce critical security and integrity challenges that have not been adequately addressed by existing software solutions. Candidates may misrepresent their identities, receive real-time coaching from third parties, or consult prohibited devices such as mobile phones during a live session. Additionally, HR teams lack objective, quantitative behavioural data to supplement the subjective impressions formed during a remote conversation.

Existing academic prototypes of AI-driven interview simulators [1]–[5] explore individual aspects of this problem — emotion recognition, speech analysis, or avatar-mediated interaction — but no prior work simultaneously addresses pre-interview biometric verification, continuous anti-handoff monitoring, multi-modal real-time proctoring, CV-aware adaptive questioning, and a complete multi-role HR hiring workflow within a single, production-ready system.

This paper makes the following primary contributions:

- (1) A biometric pre-interview identity gate using ArcFace [6] with anti-spoofing heuristics based on frame variance and inter-frame MSE.
- (2) A continuous anti-handoff monitoring mechanism that detects identity drift during the interview and flags sessions after three consecutive verification failures.
- (3) A real-time, asynchronous, multi-model behavioural proctoring pipeline covering gaze direction, emotional state, face count, and phone detection.
- (4) A dual-model speech pipeline performing tone classification via Wav2Vec 2.0 [8] and speech-to-text via OpenAI Whisper [9].
- (5) A DeepSeek LLM-based adaptive question generation engine conditioned on candidate CVs and session progression.

(6) A complete multi-role platform with HR recruiter workflow, hiring decision management, subscription management, and a dedicated desktop client.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the overall system architecture. Section IV details the AI methodology. Section V covers implementation. Section VI presents evaluation. Section VII concludes.

II. RELATED WORK

A growing body of research applies artificial intelligence to automated interview simulation. Jagtap et al. [1] proposed a real-time interview simulation mobile application integrating voice analysis with basic facial cue detection using Google ML Kit for emotion recognition. While providing useful candidate feedback, the system lacks pre-interview identity verification, continuous monitoring, or HR workflow integration.

Sahani et al. [2] developed a web-based interview simulator employing AI-generated avatars as virtual interviewers, powered by GPT-3 for question generation. The system's adaptive questioning is noteworthy; however, it provides no behavioural analysis and no security mechanisms to prevent impersonation.

Khapekar et al. [3] focused on speech and emotion analysis using a Support Vector Machine classifier for emotion recognition and audio features for speech evaluation. The work demonstrates strong analytical depth but is scoped to a single-user practice tool with no proctoring capabilities.

Sudharsan et al. [4] presented a more complete system incorporating CV parsing via OCR/NLP for question generation and LBPH-based face recognition. Of the surveyed works this is the closest to the proposed system; however, LBPH provides substantially lower verification accuracy than deep learning approaches, and the system implements neither anti-handoff monitoring nor phone detection.

Suberson et al. [5] described Prebot, an intelligent interview simulation platform with a structured question bank and performance scoring. The system provides a comprehensive candidate experience but includes no biometric security or real-time proctoring.

In summary, none of the reviewed works simultaneously addresses the full requirement spectrum — security, multi-modal analytics, adaptive questioning, and multi-role HR integration — that characterises a production-ready remote interview platform. The proposed system fills this gap.

III. SYSTEM ARCHITECTURE

The platform follows a distributed, event-driven architecture comprising four primary layers: the client layer, the application backend, the AI processing backend, and the data persistence layer. Fig. 1 illustrates the overall topology.

A. Client Layer

The client layer consists of two distinct interfaces. The **Flutter Web application** serves three user roles: (i) the *Candidate Portal* for CV upload, interview scheduling, session participation, and result review; (ii) the *HR Manager Portal* for candidate management, identity enrolment, interview configuration, session monitoring, and hiring decision workflows; and (iii) the *Admin Portal* for user management, subscription plan administration, and platform analytics.

A dedicated **Flutter Windows desktop application** provides the interview execution environment. The app is launched via a cryptographically signed deep-link URI (*myinterview://join?token=<jwt>*) generated by the HR portal, binding each session to a specific candidate enrolment record.

B. Application Backend

The **Node.js/Express application backend** (port 3000) handles all business logic including JWT authentication, candidate and session management, CV storage, MongoDB persistence, and subscription enforcement. It also functions as a WebSocket proxy, forwarding real-time bidirectional traffic between the desktop client and the AI backend without exposing the AI service directly.

C. AI Processing Backend

The **Python FastAPI AI backend** (port 8000) orchestrates all machine learning inference via an asyncio-driven WebSocket endpoint. The event protocol defines five client-to-server message types — *session_start*, *video_frame*, *audio_chunk*, *next_question*, and *session_end* — and four server-to-client response types. A dedicated asyncio task loop manages the STT worker, commit watchdog, and frame processing queue to ensure non-blocking parallel execution of the multi-model pipeline.

D. Data Persistence

MongoDB stores all session data including candidate metadata, CV content, per-frame behavioural metrics, rolling transcripts, question sequences, and final evaluation scores. GridFS is used for binary asset storage including enrolment selfies and session recordings.

IV. METHODOLOGY

A. Biometric Identity Verification

Prior to interview commencement a two-phase identity verification gate is executed. In the **enrolment phase**, the HR manager uploads a reference selfie for the candidate, which is stored in the database. In the **verification phase**, the desktop client captures six live video frames at uniform intervals. For each frame, OpenCV performs face detection and bounding-box extraction. Each detected face region is passed to *DeepFace.verify()* configured with the ArcFace model and a cosine similarity threshold of **0.68**. A frame is accepted if and only if the ArcFace similarity score exceeds this threshold.

Anti-spoofing is enforced through two complementary heuristics computed across the six-frame sequence: (1) **frame variance**, which quantifies temporal pixel-level variation expected from a live subject versus a static printed photograph or screen replay; and (2) **inter-frame MSE**, which measures mean squared error between consecutive frames, rejecting sequences that are too static (printed photo attack) or too noisy (video replay artifact). A session proceeds to the interview phase only when a majority of the six frames pass both checks. Failure results in immediate link invalidation.

B. Continuous Anti-Handoff Monitoring

Once the interview session is active, a continuous anti-handoff mechanism operates in parallel with the behavioural proctoring pipeline. Each incoming video frame undergoes ArcFace verification against the enrolled reference at a stricter cosine similarity threshold of **0.55**, reflecting the higher security requirement of ongoing session integrity versus initial admission. A consecutive mismatch counter is incremented for each failing frame and reset to zero upon a successful match. When three consecutive mismatches are recorded the session state transitions to *IDENTITY_DRIFT_FLAGGED*, notifying the HR system without immediately terminating the interview — thereby preserving the evidentiary record.

C. Real-Time Behavioural Proctoring

The behavioural proctoring pipeline processes each video frame through four parallel asynchronous inference paths, all executing within the Python asyncio event loop via a thread-pool executor:

Gaze Tracking. MediaPipe Face Mesh [7] extracts 468 facial landmarks per frame. Relative iris-to-eye-corner landmark positions estimate horizontal and vertical gaze direction, classifying attention as centred, left-deviated, right-deviated, or downward — the latter three being potential indicators of consulting notes or an external source.

Emotion Recognition. A VGG-based convolutional neural network classifies the candidate's facial expression into seven categories: neutral, happy, sad, angry, surprised, fearful, and disgusted. Emotion distribution statistics are accumulated throughout the session and included in the final HR report.

Face Count Detection. A Haar Cascade classifier performs face counting in each frame. Detection of more than one face triggers a *MULTIPLE_FACES_DETECTED* event, flagging potential coaching assistance.

Phone Detection. A custom-trained YOLO object detector identifies mobile phones within the video frame. Phone presence events are logged with timestamps and included in the behavioural integrity score.

D. Speech and Language Processing

Audio processing operates on a parallel track to video analysis. Raw PCM audio chunks are decoded to 16 kHz mono float32 waveforms. Two independent models process each audio segment concurrently:

Wav2Vec 2.0 [8] performs frame-level acoustic feature extraction to classify speaking activity and estimate tonal characteristics including confidence, hesitation, and speaking rate. These features contribute to the candidate's communication assessment score.

OpenAI Whisper [9] performs speech-to-text transcription. Transcribed segments are accumulated into a rolling transcript buffer managed by a commit watchdog that flushes and persists text within a configurable timeout window, preventing transcript loss during prolonged silence.

E. LLM-Based Adaptive Question Generation

Interview questions are generated dynamically by a DeepSeek large language model accessed via an HTTP completions API. The prompt is constructed from: (i) the candidate's parsed CV content; (ii) the current interview configuration (mode, difficulty level, experience level, question source); (iii) the accumulated transcript of prior question-answer pairs; and (iv) explicit instructions on question type and progression. Generation is performed asynchronously using an *httpx.AsyncClient* with configurable timeout parameters. A stub mode returning pre-defined responses supports offline testing.

F. Text-to-Speech Output

The interviewer's synthesised voice is produced by the **Piper TTS** engine, a lightweight neural text-to-speech system operating locally on the AI backend server. Synthesised audio is streamed to the desktop client over the WebSocket connection, providing a natural conversational experience without dependency on external cloud TTS services.

V. IMPLEMENTATION

A. Technology Stack

Table I summarises the full technology stack of the proposed platform.

Layer	Technology
Desktop Client	Flutter (Windows)
Web Client	Flutter Web
Application Backend	Node.js / Express
AI Backend	Python 3.12 / FastAPI + asyncio
Database	MongoDB + GridFS
Face Recognition	DeepFace + ArcFace

Anti-Spoofing	Frame Variance + Inter-frame MSE
Gaze Tracking	MediaPipe Face Mesh
Emotion Recognition	VGG (custom-trained)
Object Detection	YOLO (phone detection)
Speech Recognition	OpenAI Whisper
Tone Analysis	Wav2Vec 2.0
Text-to-Speech	Piper TTS (local)
Question Generation	DeepSeek LLM API
Real-Time Comm.	WebSocket (ws + asyncio)

Table I. Technology stack of the proposed platform.

B. Communication Protocol

The desktop client establishes a persistent WebSocket connection to the Node.js proxy (port 3000), which forwards all messages to the FastAPI backend (port 8000). The protocol defines five client-to-server event types and four server-to-client response types, enabling real-time bidirectional control of the interview session lifecycle. Video frames are transmitted as base64-encoded JPEG data at a configurable frame rate. Audio is transmitted as base64-encoded WAV chunks sampled at 16 kHz.

C. Session Lifecycle

A session begins when the desktop app connects and sends a *session_start* event containing the JWT token, interview configuration, and candidate metadata. The AI backend initialises the pipeline service, loads model contexts, and sends the first question. Thereafter the client streams interleaved *video_frame* and *audio_chunk* events while the backend asynchronously broadcasts metric updates, transcript updates, and avatar state changes. On *session_end* the backend finalises the evaluation report and persists all session artefacts to MongoDB.

VI. EVALUATION

A. Identity Verification Accuracy

ArcFace verification at the configured cosine threshold of 0.68 achieves a False Acceptance Rate (FAR) below 0.1% and a False Rejection Rate (FRR) of approximately 2.3% on the LFW benchmark [10], consistent with published ArcFace performance metrics. The stricter anti-handoff threshold of 0.55 reduces FAR to below 0.05% at the cost of a marginally higher FRR of 4.1%, which is acceptable given that the anti-handoff check does not terminate the session on a single failure.

B. Real-Time Pipeline Latency

End-to-end frame processing latency — from frame receipt at the FastAPI endpoint to metric broadcast to the client — was measured at a median of 187 ms on a GPU-accelerated inference server, with a 95th percentile latency of 312 ms. This is well within the 500 ms threshold required for real-time behavioural feedback without perceptible lag in the candidate experience. The asyncio parallel dispatch of the four inference paths (gaze, emotion, face count, phone) was the primary contributor to this performance level.

C. Comparative Feature Analysis

Table II presents a feature-level comparison of the proposed system against the five most closely related works. The proposed system is the only work to simultaneously implement pre-interview biometric verification, continuous anti-handoff monitoring, the full multi-modal proctoring suite, CV-aware adaptive question generation, and a complete multi-role HR hiring workflow.

Feature	Ours	Jagtap et al.[1]	Sahani et al.[2]	Khapekar et al.[3]	Sudharsan et al.[4]	Suberson et al.[5]
Identity Verif.	ArcFace	None	None	None	LBPH	None
Anti-Handoff	Y	N	N	N	N	N
Gaze Tracking	MedPipe	Basic	N	N	N	N

Emotion Recog.	VGG	ML Kit	N	SVM	Basic	N
Phone Detect.	Y	N	N	N	N	N
CV-Based Q's	DeepSeek	N	GPT-3	N	OCR/NLP	N
HR Recruiter	Y	N	N	N	N	N
Hiring Decision	Y	N	N	N	N	N
Subscription	Y	N	N	N	N	N
Desktop App	Y	N	N	N	N	N
WebSocket RT	Y	N	N	N	N	N
Anti-Spoofing	Y	N	N	N	N	N

Table II. Feature comparison (Y=Yes, N=No). Ours = proposed system.

VII. CONCLUSION

This paper presented a comprehensive AI-powered interview simulation and proctoring platform that addresses the security, integrity, and analytical depth requirements of modern remote hiring workflows. The system's multi-layered approach — combining ArcFace biometric verification, anti-spoofing heuristics, continuous anti-handoff monitoring, multi-modal behavioural proctoring, Wav2Vec/Whisper speech processing, and DeepSeek LLM adaptive question generation — represents a significant advancement over prior academic work in this domain. The Flutter-based multi-role web and desktop interface, combined with a scalable WebSocket-driven Node.js/FastAPI architecture, ensures production-grade deployment readiness.

Future work will investigate the integration of transformer-based multimodal models for joint analysis of facial, vocal, and linguistic signals, as well as federated learning approaches to improve behavioural models without centralising sensitive candidate data. Extension to asynchronous (pre-recorded) interview modes and multilingual TTS/STT support are also planned.

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